# QUASI-COHERENT MODULES ARE DIRECTED COLIMITS OF FINITE TYPE SUBMODULES

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*Editorial note.* Counted span is the author's own proof environment, source lines 2527-2540, following the statement at 2518-2525.

*Editorial note.* No omissions.

*Allowed standard prerequisites (editorial).*

- Colimit of finitely presented modules. lemma-colimit-finitely-presented (same chapter, cited at ~2530-2533). F = colim F_i with F_i finitely presented quasi-coherent.

*Source reference.* Original lemma 07V9: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2492–2492. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be a quasi-compact and quasi-separated algebraic space over $S$. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then $\mathcal{F}$ is the directed colimit of its finite type quasi-coherent submodules.*

*Proof.* If $\mathcal{G}, \mathcal{H} \subset \mathcal{F}$ are finite type quasi-coherent $\mathcal{O}_X$-submodules then the image of $\mathcal{G} \oplus \mathcal{H} \to \mathcal{F}$ is another finite type quasi-coherent $\mathcal{O}_X$-submodule which contains both of them. In this way we see that the system is directed. To show that $\mathcal{F}$ is the colimit of this system, write $\mathcal{F} = \operatorname{colim}_i \mathcal{F}_i$ as a directed colimit of finitely presented quasi-coherent sheaves as in Lemma 07V9. Then the images $\mathcal{G}_i = \operatorname{Im}(\mathcal{F}_i \to \mathcal{F})$ are finite type quasi-coherent subsheaves of $\mathcal{F}$. Since $\mathcal{F}$ is the colimit of these the result follows. □

*Date*: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.